# STRATA: A Process-Mining Framework for Detecting and Locating HVAC Faults from Building Trend Data

Yassh Singh
Thompson Rivers University, Kamloops, Canada
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Supervisors: Dr. Anthony Aighobahi and Dr. Mridula Sharma
singhy21@mytru.ca

Abstract—Faults and poor control in heating, ventilation and air-conditioning (HVAC) equipment waste 15 to 30 percent of a commercial building's energy, and many go unnoticed. Building automation systems log the operating data and display it as trend graphs, so a fault is found only when a technician reads the graphs. Automated tools either apply rules written in advance, which miss any fault that was not anticipated, or report anomaly scores that say a day is unusual without saying where or what is wrong. This paper presents STRATA, a framework that works from the trend data a building already records and learns, from one period of normal, fault-free operation, what each day should look like. It then examines every new day with eight checks, each asking whether a physical relationship or an operating sequence has moved outside its normal range. Each check is first run on fault-free days it has never seen. The number of those days it wrongly flags is its false-alarm rate. A check may then raise an alarm only when it flags far more days than that false-alarm rate could explain. Every alarm states what broke, where, the evidence and the check's measured false-alarm rate. Process mining supplies the structure. The sensor data becomes a log of events along the control sequence, and models of normal operation are discovered from that log for each zone's equipment and for the whole unit, so every alarm knows which piece of equipment it belongs to. On five public benchmark systems from Lawrence Berkeley National Laboratory, each simulated with faults whose cause and location are known, STRATA detected 158 of 175 faults, most on the first day, and raised false alarms on 26 of 480 fault-free days it had never seen. In the 50 cases where the fault lay inside one zone's equipment, the alarm pointed to the correct zone in 37, to no zone in 13, and to a wrong zone in none. Moving the framework to a new building means writing one configuration file, not new code. The planned next step is a live trial on a real building. It is also the first step toward ProcessHeal, a self-healing building that finds its own faults, locates them, works out their cause and corrects them. STRATA does the finding and the locating; working out the cause and correcting it are the later stages.

Index Terms—HVAC, fault detection, process mining, building automation, event logs, explainable alarms

## I. Introduction

Buildings account for about a third of energy-related carbon emissions [1], and HVAC accounts for up to half of a building's energy use [2]. Faults and poor control that go uncorrected waste 15 to 30 percent of a commercial building's energy [3]. A study of more than 60,000 pieces of HVAC equipment found 40 percent of air handlers in a faulted state on any given day, in buildings that already had automation systems with alarms [4]. Seven of the ten most common air-handler faults in that study are faults of behaviour, such as a wrong schedule or an unreleased manual override, rather than a broken part.

A building operator today usually learns of such a fault in one of three ways. An occupant complains that a room is too hot or too cold. The operator notices that a line on a trend graph has drifted from where it usually sits. Or an automated detector raises an alarm. The first two find the fault late and depend on someone happening to look. The third works only as well as the method behind it.

Those automated methods fall into two groups. Rule libraries, which come built into commercial building automation systems, are transparent and need no training data, but they catch only the faults for which someone wrote a rule. Machine-learning detectors learn from the building's own data and so can find faults that no rule anticipated, but most report an anomaly score: a number that says today is unusual without saying where or what is wrong. A score is not enough to write a work order.

This paper presents STRATA. It is configured with rules, written once per building in a single file, but those rules do not raise alarms. They turn the sensor data into a log of events: cooling started, damper open while commanded closed, a supply-air reading that physics says cannot be right. What a normal day of those events looks like is then learned from a period of the building's own fault-free operation, and every new day is checked against it, so a fault can show up as a departure from normal even when no rule names it. Each alarm states the fault, the piece of equipment, the evidence, and how often that same check has been wrong on fault-free days.

The method behind it is process mining, which discovers a model of a process from a log of events and checks later behaviour against the model [5]. STRATA builds such a log from the sensor data, for each zone's equipment and for the whole unit. That is how an alarm knows where it belongs, and why it can check the order of control actions as well as physical relationships. To our

knowledge this is the first use of process mining for HVAC fault detection: the surveys of both fields contain no such entry, and our own search found none (Section II). On five public benchmark systems with 175 seeded faults, STRATA detected 158, most on the first day, and placed no alarm in a wrong zone (Section IV). The code and the complete evaluation are public.[1]

## II. Related Work

### A. Rule-based detection, the method in use today

Fault detection in real buildings is almost entirely rule-based. The air-handling-unit performance assessment rules of Schein and colleagues set the pattern [6]: a rule states a relationship that must hold in a given operating mode, such as no water flow through a closed valve, and fires when it does not. ASHRAE Guideline 36 publishes such fault conditions alongside its control sequences [7], [8], an open-source implementation of them exists [?], a survey of fourteen commercial fault-detection tools found this rule-and-threshold style to be the norm [9], and interviews with 29 industry experts in 2024 found it still so [10]. Installed is not the same as used: in a two-site field trial of Guideline 36, the facility team paid no attention to the built-in fault detection [11], and a 2026 California code study names alarm overload and a shortage of trained staff as the main barriers [12]. When alarms outnumber the people who can act on them, false alarms erode trust in the true ones [13]. The Lawrence Berkeley National Laboratory group has carried the rule approach through to correction, field-testing seven hand-coded correction algorithms on 225 pieces of equipment in four buildings [14], [15]. Rules work from the first day and need no baseline data. Their limit is structural: they catch only the faults someone wrote a rule for, and miss the gradual, multi-variable and sequence faults that no rule author anticipated. The LBNL group itself notes that commercial diagnostics often list several candidate causes or none [16], and recent work answers that with active tests that command the equipment and watch the response [17]. Section IV measures both the strength and the limit against the Guideline 36 rules on the same benchmarks.

### B. Machine learning, and what its alarms leave out

A 2024 systematic review covers 211 studies of artificial intelligence for HVAC fault detection [2], and a 2025 review follows the field through the equipment lifecycle [18]. Accuracy on the standard benchmarks has saturated: 98 percent across thirteen fault types with a random-forest and support-vector pipeline [19], 97 percent on data from three real buildings with a convolutional and recurrent network [20]. Two problems remain. The first is what the output is. A classifier or an anomaly score says that something is wrong; where an explanation is offered it comes from a separate layer that ranks input features, as in the graph network that reaches 88 percent on one of the benchmarks used here with per-sensor importance as its explanation [21], [22]. That answers which signal mattered, not what the equipment should have been doing. The closest machine-learning work to a located, evidenced alarm is recent: a model trained only on fault-free data that estimates the position of a stuck damper or valve [23], and a Bayesian network that turns residuals into evidence and isolates the faulty component without labels [24]. The second is labels. Real buildings have no fault labels, so the strongest recent work is unsupervised: self-supervised transformers [25] and autoencoders trained on a Canadian university's building automation data [26], each of which yields a scalar deviation score. Moving a learned model from one building to another is a research problem of its own [27], [28]: in a 2026 study, detectors that scored 0.8 F1 inside their own buildings fell to 0.45 on unseen ones [29]. A live deployment on a Danish school building weighted its choice of detector toward avoiding false alarms [30], and an audit of 65 such studies found two that shared code [31]. The newest direction loops back toward rules: a large language model that writes and refines detection rules under physical constraints, and must report its evidence, beats the neural detectors on the LBNL air-handler data [32], although a 2026 review of 66 such studies found none in sustained deployment [33].

### C. Process mining, and how close it has come to buildings

Process mining discovers a model of a process from an event log and checks later behaviour against that model [5]; the inductive miner used here is its standard discovery algorithm [34]. Checking conformance has served as an anomaly detector on event logs for a decade [35]. Applying it to sensor data needs one extra step, event abstraction, which turns continuous signals into discrete events [36], [37]; how to do that well, and how to keep the resulting events checkable by a person, is an open question in the process-mining community as of 2026 [38], [39]. With that step the method has reached several physical systems: anomaly detection in chemical plants [40] and in industrial alarm sequences [41], digital twins that discover normal behaviour offline and check conformance online [42], fault diagnosis on a robotic arm with an F1 score up to 98.9 percent [43], process-mining features that beat plain conformance checking by 30 to 70 F1 points on four datasets [44], and sensor validation in industrial Internet-of-Things systems [45]. The closest analogue to a building is pasteurisation, where sensor signals are abstracted into activities and checked against the critical control points of the HACCP food-safety standard [46]; the parallel between those control points and Guideline 36 control sequences is the one STRATA builds on. Petri nets of HVAC equipment drawn by hand also exist [47], but a net an engineer draws is not a model discovered from a log, and that distinction is the whole method.

[1] https://github.com/iyassh/strata

### D. The gap, and where STRATA sits

No published work applies process mining to building HVAC fault detection. The 211-study review has no process-mining entry in its taxonomy of 23 methods [2]; a review of 47 process-mining studies in Industry 4.0 has no building application [48]; a review of 36 process-mining studies on sensor data covers industry, smart homes and healthcare, with no HVAC entry [37]; and the LBNL taxonomy of fault-detection methods does not list it [9]. A title-and-abstract search of the OpenAlex index for process mining together with HVAC, air handling, building automation, building energy or chillers, run on 16 October 2026, returns no work since 2023, and the programmes of the process-mining conferences and workshops of 2024 to 2026 contain none. The nearest work analyses building-management alarm logs statistically, without discovering a process model from them [49]. STRATA therefore sits between the two established groups. It learns from fault-free data what the building's normal operation is, which a rule library cannot, and it reports a location, the evidence and a measured false-alarm rate, which a score cannot. It is a complement to a rule library rather than a replacement, and its located alarms are the kind of input the LBNL correction algorithms were built to act on.

## III. The Framework, Stage by Stage

Figure 1 shows the framework as four tiers. The models and thresholds in the top two tiers are built once per building from one fault-free period; the bottom two tiers, and the event conversion of Stage 2, run on every new day. The stages below follow the figure from top to bottom. Table I defines the terms they use.

TABLE I
Terms used in this paper.

| Term | Meaning |
|---|---|
| Trend data | The minute-by-minute sensor and actuator values a building-automation system records. |
| Unit | The equipment scored as a whole: an air handler, a rooftop unit or a fan coil unit. |
| Zone | A room or group of rooms served by one terminal unit. |
| Terminal unit | The box that serves one zone: its damper, reheat valve and, in some designs, a small fan. |
| Event | One thing that happened at one time, such as "cooling started at 09:14". |
| Event log | All the events of all the days, one case per day. |
| Process model | A graph, discovered from the log, of the states a terminal unit or a unit passes through on a normal day and in what order. |
| Conformance | How well one day's events fit the process model. |
| Healthy band | The range a value or a daily count stayed within on the fault-free training days. |
| Check | One of the eight tests run on each day; seven can raise an alarm, the eighth only supports one. |
| Seeded fault | A fault written into a simulation on purpose, so the right answer is known. |
| Scenario | One simulated run, usually a year, that contains one seeded fault. |
| Held-out days | Fault-free days kept aside and used only to measure how often each check gives a false alarm; the remaining fault-free days are the training days. |
| Measured false-alarm rate | The false-alarm rate of a check, measured on the held-out days, and reported with every alarm the check raises. |

### A. Stage 1: the building's data and one configuration file

The input is the trend data a building-automation system already records: temperatures, flows, pressures, damper and valve positions, fan speeds, usually once a minute. STRATA does not need new sensors. It needs one fault-free period of that data, ideally a year so that every season is seen, and one configuration file.

The configuration file is the only thing that is specific to a building. It has two parts. The first maps the building's sensor names to the roles the framework understands: for example, which column holds the supply-air temperature and which holds each zone's damper position. The second lists the control sequence for this type of equipment as rules, taken from the design documents, from the ASHRAE Guideline 36 sequences [7] and, as Stage 3 describes, from invariants found in the fault-free data. These rules come in two families. State rules say when the unit is occupied and when it is in heating or cooling mode. Signature rules say which relationships must hold, such as no flow through a closed valve.

Onboarding a new building means writing this file; the framework's code does not change. On the benchmarks, once the framework existed, configuring a second equipment type took about one and a half hours. Two of the five main benchmark systems were configured before any fault data was opened, so nothing in them could have been tuned to the faults.

### B. Stage 2: turning sensor data into events

Process mining works on event logs, not on sensor curves, so the first thing the framework does with the data is turn the minute-by-minute values into events along the control sequence. Each day is one case. An event is, for example, "fan started at 06:02" or "cooling became active at 09:14". The configuration's rules produce them: a state rule for operating mode fires an event when the mode changes, and a signature rule fires one when the relationship it guards is broken.

There are two kinds of events, and keeping them apart is a central design decision. State events describe what the equipment is doing, without any judgement of whether it is correct. Signature events describe something that should not happen, such as "water flowing through a valve that

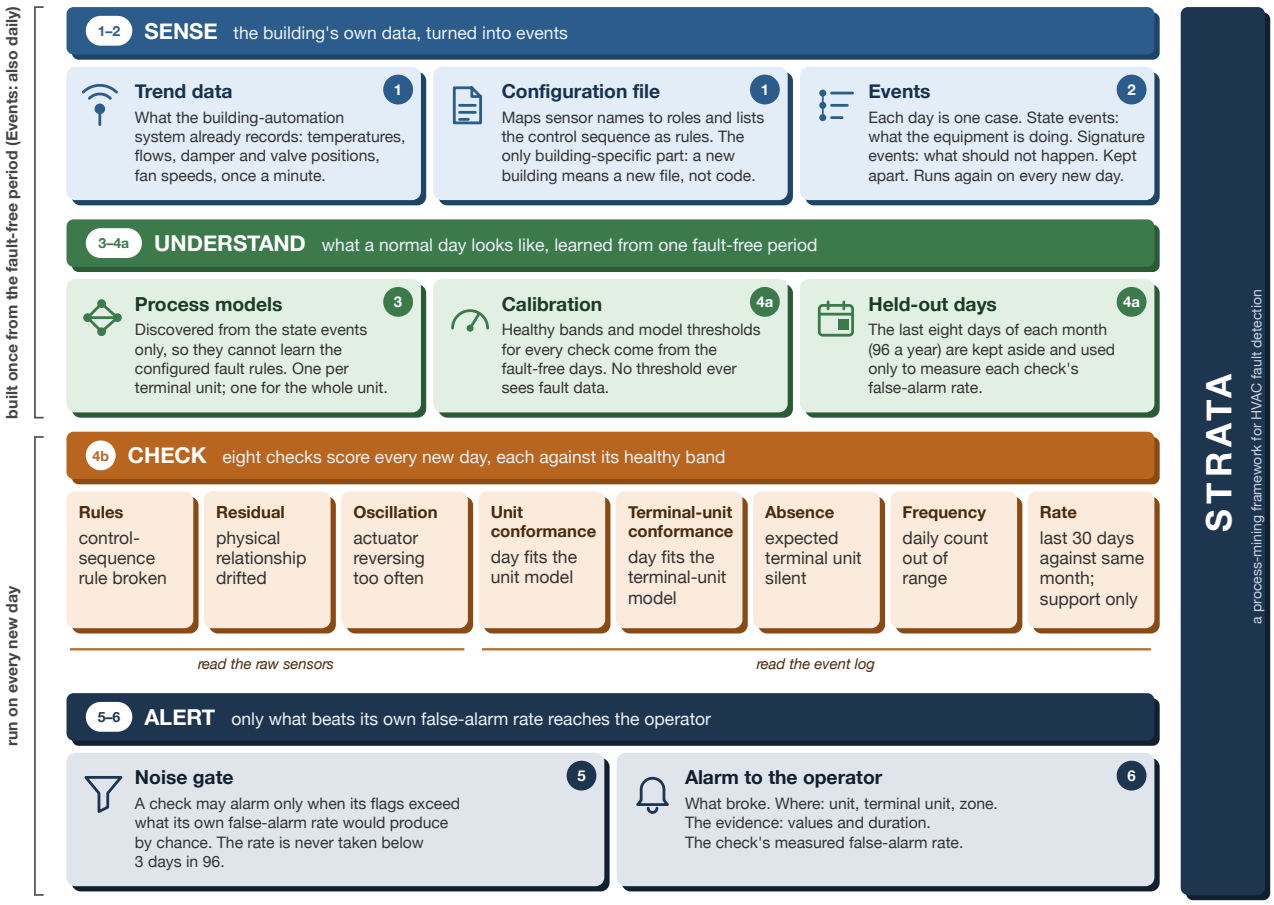



Fig. 1. The STRATA framework as four tiers. The top two tiers are built once per building from one fault-free period: the trend data is turned into events, process models of a normal day are discovered from the state events, and the eight checks are calibrated, with their false-alarm rates measured on held-out days. The bottom two tiers run on every new day, after the day has been turned into events by the same Stage 2 rules: the eight checks score it, the noise gate passes only alarms that beat the check's own false-alarm rate, and what passes reaches the operator stating what broke, where, the evidence and that rate. The numbered badges are the stages of Section III; 4a (calibration) and 4b (the checks) are the two halves of Stage 4.

is commanded closed". Only the state events are allowed into the next stage, model discovery. If the signature events were allowed in, the discovered model would simply learn the configured fault rules and later "detect" them, which would be circular. We measured the effect: on one system, letting signature events into discovery raised the unit conformance check's apparent detection from 1.8 to 5.2 percent of fault days, and every added day was the model re-finding a rule we had written.

### C. Stage 3: discovering what normal looks like

From the state-event log of the fault-free period, the framework discovers process models of a normal day with a standard process-mining algorithm, the inductive miner [34]. It does this at two levels; the levels are the strata the name refers to. The terminal-unit level has one model per terminal unit, describing the normal day of that terminal unit's damper and reheat valve. The unit level has one model for the unit's whole daily cycle (the air handler, in the systems that have terminal units). Each model is a graph of the states the terminal unit or unit passes through and the order it passes through them (a Petri net, in process-mining terms).

This structure does two things in the framework. First, it gives every alarm its location: events, rules and models are all defined per terminal unit, so any check that fires on one terminal unit's data names that terminal unit, and a check that fires on the air handler's data names the air handler. Second, during development the models surfaced invariants of normal operation, things that were always true on fault-free days, which we then wrote down as rules in the configuration. The models do not detect faults by themselves; Section IV-D reports the test that established this.

### D. Stage 4: calibration, then eight checks

Calibration (4a). Every threshold in every check comes from the fault-free period only. The year is split by

calendar date: the last eight days of each month (96 days in a year) are held out and never used for setting anything; they exist only to measure how often each check gives a false alarm. The remaining days, the training days, set the healthy bands and train the models. No threshold ever sees fault data, and no threshold sees the days on which its false-alarm rate is later reported.

The checks (4b). Each day is scored by eight checks, each of which alarms on its own and is named in the alarm it raises. Three of them read the raw sensor data:

- Rules: a control-sequence rule was broken, such as flow through a closed valve, or a zone's airflow far from its setpoint while occupied.
- Residual: a physical relationship that must hold in a known mode has drifted out of its healthy band. With the cooling coil off, the supply-air temperature should equal the mixed-air temperature plus the small rise from the fan; if it does not, a sensor is biased or a coil is leaking.
- Oscillation: an actuator reversed direction far more often than it does on fault-free days, the sign of an unstable control loop.

Five of them read the event log:

- Unit conformance and terminal-unit conformance: how well today's sequence of events fits the discovered model of the unit or of the terminal unit. A day fits poorly when, for example, cooling starts before the fan, an order the fault-free period never showed.
- Absence: a terminal unit that produces events on every fault-free occupied day produced none on an occupied day.
- Frequency: a daily count, such as the number of heating episodes, left its healthy band.
- Rate: the number of the last 30 days on which a mode was active, compared with the same calendar month of the fault-free year. It never raises an alarm alone, because on one system it added false alarms; it is reported beside another check's alarm as corroborating evidence.

### E. Stage 5: the noise gate

A check is allowed to raise an alarm on a scenario only when the number of days it flagged is too large to be explained by its own false-alarm rate. The test is a binomial test, the same calculation used for coin tosses. Suppose a check wrongly flags three of 96 held-out fault-free days. Flagging forty of 365 days in a scenario is then far beyond what chance would give, and the check may alarm; flagging five is within chance, and the check may not alarm. For the test, the rate is never taken below three days in 96, so a check with no false alarms on the held-out days is not treated as perfect. The alarm itself reports the measured count, which can be zero, as in Figure 2.

On the benchmarks the test is applied once per check per scenario, to decide whether the check is credited with the detection; the first day the credited check flagged is the day of the alarm. On a live building there is no scenario, only days. How the test is applied there, for example over a rolling window of recent days, is a decision for the planned trial (step 4 of Table IV). What is already fixed is that every check's false-alarm rate is measured on the held-out days before the first alarm and is reported with each alarm, so the operator knows how often that check raises a false one.

```
STRATA alarm -- series fan-powered terminal
                units, 2018-01-01 (day 1 of fault)
 what       zone primary-air flow not tracking
          its setpoint            [check: rules]
 where       terminal unit TU_S, zone S
 evidence   715 CFM measured against a 200 CFM
          setpoint, a 515 CFM excess, held for
          720 min while occupied; damper 0.80
          open, 0.32 on the same date of the
          fault-free year
 rate        rules check: 0 false-alarm days on
          the 96 held-out fault-free days
 ---------------------------------------------
 truth       zone-S damper stuck at 80%: location
          and symptom correct, caught on day 1
```

Fig. 2. One alarm from the series fan-powered benchmark. Every value is taken from the framework's recorded output and the benchmark data; the last line is the answer key, not part of the alarm. The alarm names the symptom and the location; it does not claim the root cause.

### F. Stage 6: the alarm the operator receives

What survives the gate reaches the operator with four things: what broke, where, the evidence, and the check's measured false-alarm rate. Figure 2 shows one, assembled from the framework's recorded output. The fault seeded was a zone damper stuck at 80 percent open. The alarm names the zone, names the terminal unit, states that the primary airflow is 515 cubic feet per minute (CFM) above its setpoint and has been for twelve hours while occupied, reports the damper position of 0.80 against a normal 0.32, and tells the operator that the rules check gave zero false alarms on the 96 held-out fault-free days. It was raised on the first day of the fault.

## IV. Results on Public Benchmark Data

We evaluated the framework on the public HVAC fault-detection datasets from Lawrence Berkeley National Laboratory [50]. Each simulated dataset is one-minute data for one type of equipment, recorded once fault-free and once per seeded fault: a stuck damper, a biased sensor, a fouled coil, a leaking valve, an unstable controller. The five main systems have a full year of data.

Table II lists the seven datasets: air handlers that serve many zones through one duct (single-duct) or through separate hot and cold ducts (dual-duct), two kinds of fan-powered terminal unit, a fan coil unit that serves one room, and two rooftop units. Six are simulated; one is a measured year from a real rooftop unit. We found no published detection results on the fan-powered or dual-duct datasets. Unless stated, the results below are for

TABLE II
The seven LBNL datasets. "Scored" is the number of seeded faults evaluated; "Detected" those the framework caught; "False-alarm days" the held-out fault-free days on which it alarmed. †One of these was attributed on audit to a dataset defect rather than to the fault; the totals in the text count it, and are 157 without it. The rooftop datasets are shorter, hence fewer held-out days.

| System | Scored | Detected | False-alarm days |
|---|---|---|---|
| Single-duct air handler | 14 | 14† | 1 of 96 |
| Fan-powered terminal units (parallel) | 30 | 26 | 8 of 96 |
| Fan-powered terminal units (series) | 29 | 25 | 9 of 96 |
| Dual-duct air handler | 55 | 50 | 3 of 96 |
| Fan coil unit | 47 | 43 | 5 of 96 |
| Rooftop unit (simulated) | 24 | 20 | 0 of 28 |
| Rooftop unit (measured) | 1 case | — | 3 of 49 |

the five main systems. The sixth dataset, a simulated rooftop unit, is the only one that records refrigerant-side measurements, so it was used for the faults that are visible only on that side. The measured unit has 182 fault-free days and one fault case, so it yields a false-alarm rate but no detection count.

## A. Detection and false alarms

Across the five main systems the framework detected 158 of 175 seeded faults, with 26 false-alarm days in 480 held-out fault-free days, between 1 and 9.4 percent per system. The median time from the start of a fault to its first alarm was one day. On the real rooftop unit the false-alarm rate was 3 of 49 held-out days. A later audit of the configurations mapped every recorded column that the first configurations had left unread; sixteen of the 158 detections came from columns that audit added. The 17 faults it missed are all coil fouling. For some coils every seeded severity was missed, because the recorded sensor values stay inside every healthy band; nothing that reads only those values could see them. On the one dataset that records the refrigerant side, the framework caught every fouling severity.

## B. Where the alarm points

On the two fan-powered systems, where each air handler serves four terminal units, 50 detections made before the column audit had their seeded fault in one terminal unit. In 37 of these the alarm named the correct terminal unit, in 13 it named the air handler but no terminal unit, and in none did it name a wrong one. Over all 158 detections the rule carrying the alarm pointed at the right class of component (coil, fan, damper or terminal unit) in 83 percent of cases and at the exact seeded component in 69 percent. The alarm names the symptom and the place. It does not name the root cause: a first diagnosis layer we tested improved localisation slightly but did not reach the targets we had set for it, so we make no diagnosis claim.

TABLE III
STRATA against the Guideline 36 fault rules on the three systems on which both were run, under the same gate. "Detected" is seeded faults detected; "False" is false-alarm days among the held-out days. The single-duct count is the audited 13 of Table II.

| | STRATA | | Guideline 36 rules | |
|---|---|---|---|---|
| System | Detected | False | Detected | False |
| Single-duct air handler | 13 of 14 | 1 of 96 | 12 of 14 | 1 of 96 |
| Fan-powered terminal units (parallel) | 26 of 30 | 8 of 96 | 8 of 30 | 0 of 96 |
| Fan-powered terminal units (series) | 25 of 29 | 9 of 96 | 9 of 29 | 0 of 96 |

## C. Against two baselines

Against principal component analysis, a standard statistical method applied to the raw sensors, the framework detected 64 of 73 shared scenarios against the baseline's 61, a difference too small to be statistically significant. The baseline cannot name the terminal unit. Against the Guideline 36 fault rules that commercial systems run, implemented by an independent open-source library and scored under the same gate, the framework detected one more fault than the rules on the single-duct air handler at the same false-alarm count, and three times as many on the terminal units, because it reads the per-zone sensors the rules do not (Table III). We had predicted that STRATA would detect at least as many faults as the rules at a lower false-alarm rate on every system. It did not: on the air handler the false-alarm counts were equal, and on the terminal units the extra coverage cost eight and nine false-alarm days against the rules' none.

## D. What process mining contributes

We began with the hypothesis that the discovered process models would detect the faults themselves. They did not. Over 199 scenarios on six systems (the 175 plus the 24 on the simulated rooftop unit), no fault was caught by the conformance checks alone, and even on the scenario where the models did best, a one-line hand-written rule flagged more fault days than the model did. What process mining contributes is structure. The per-terminal-unit models give every alarm its location. The models surfaced invariants that became detection rules, for example that the air handler's heating never runs on an occupied workday. On the two fan-powered systems the frequency check, which counts events in the log, is one of only two checks (with the oscillation check) that see the unstable-controller faults. Process mining helps to get detection but is not detecting on its own. That finding came from pre-registered tests, with the refuting result written down before each one was run, and it is why detection rests on the calibrated checks while the models supply structure and location.

### E. Robustness to noisier data

Simulated data is cleaner than a real building's. To check that this had not inflated the results, we applied four perturbations to the fault-free year and every fault file: sensor noise; change-of-value logging, which records a value only when it changes, as many real systems do; gaps in the record; and shifted schedules. We then re-fitted and re-scored all five systems. Before running this test we had set a limit of 10 false-alarm days in 96. Every system stayed under it, with the worst count at 7 of 96, and each condition cost at most one detection per system.

## V. Deployment on a Real Building: The Next Step

This section describes planned work; none of it has been carried out. Thompson Rivers University (TRU) is the example because the work was done there and a trial is planned. A 2022 energy study of the Campus Activity Centre lists an air-handling unit serving variable-air-volume terminal units, fan coil units and five rooftop units [51]. All are equipment types on which the framework has been evaluated, and nearly all are logged by one Automated Logic building-automation system. Table IV gives the steps.

Three things decide whether it works on a real building, and all three are checked before the first alarm is trusted. First, the fault-free period must really be fault-free: one faulty day in it can move a band edge, so the period is audited first. Second, the configuration must read every recorded point; the column audit of Section IV showed what unread columns cost. Third, the silence step must run at the building's own sampling rate, because thresholds set on one-minute data do not transfer to fifteen-minute data.

TABLE IV
Deploying the framework on a building. Steps 1 to 3 are the top two tiers of Figure 1, set up once; step 4 is the bottom two, with the event conversion of Stage 2, run every day.

| Step | What is needed | What comes out |
|---|---|---|
| 1. Collect | A fault-free period of trend data from the building-automation system at its own sampling rate; a year is best | The fault-free baseline |
| 2. Configure | The point list (every sensor and actuator the system records), mapped to the framework's roles; the control sequence as rules; every recorded point mapped or excluded with a reason | The configuration file, the only file specific to the site |
| 3. Confirm silence | Run the signature rules on the whole fault-free period; any rule that fires there is fixed or removed. The other checks are then calibrated on the training days, and the held-out days give every check's false-alarm rate | A detector whose false-alarm rates are measured on this building's held-out days |
| 4. Operate | Each day's data, turned into events, scored by the eight checks and passed through the noise gate; the window over which the gate counts flagged days is set in the trial | Alarms stating what broke, where, the evidence and the check's measured false-alarm rate |

## VI. Limitations

Every detection result above comes from simulated data, and simulated years are cleaner than real ones; the one real building so far supports a false-alarm rate, not a detection rate. Two scoring choices were made after we saw their effect. The rate check was demoted to support-only, and the two conformance checks, whose thresholds had been set on the same days later used to measure their false-alarm rates, were re-set on separate days, which raised the unit conformance false-alarm rates on the two fan-powered systems. Slowly developing faults are caught late, up to 114 days for two fouling cases. The benchmarks seed equipment and sensor faults, not the schedule and override faults that are most common in real buildings, such as a unit left in manual or a schedule left in holiday mode. Finally, the alarm names the symptom and the location, not the cause; diagnosis and automated correction are the later stages of ProcessHeal, and would need active tests and an operator study before any automated action.

## VII. Conclusion

STRATA turns the trend data a building already records into alarms an operator can act on: it learns normal operation from one fault-free period, checks each day against physics and against discovered process models, lets through only alarms that beat a measured false-alarm rate, and states what broke, where, the evidence and that rate. On the five main benchmark systems it detected 158 of 175 seeded faults, with the false-alarm rate of every check measured on held-out days and reported with each alarm, and, in the 50 cases where this could be checked, never named a wrong terminal unit. It is onboarded with a configuration file rather than new code. Process mining gives the framework its structure and the location on every alarm; the detecting is done by the calibrated checks. The next step is a live trial on a real building, and the planned site is the TRU campus.

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## References

[1] United Nations Environment Programme and Global Alliance for Buildings and Construction, "Global status report for buildings and construction 2024/2025: Not just another brick in the wall," United Nations Environment Programme, Nairobi, Tech. Rep., 2025. [Online]. Available: https://globalabc.org/resources/publications/global-status-report-buildings-and-construction-20242025-not-just-another

[2] J. Bi, H. Wang, E. Yan, C. Wang, K. Yan, L. Jiang, and B. Yang, "AI in HVAC fault detection and diagnosis: A systematic review," Energy Reviews, vol. 3, no. 2, p. 100071, 2024.

[3] S. Katipamula and M. R. Brambley, "Methods for fault detection, diagnostics, and prognostics for building systems—a review, part I," HVAC&R Research, vol. 11, no. 1, pp. 3–25, 2005.

[4] E. Crowe, Y. Chen, H. Reeve, D. Yuill, A. Ebrahimifakhar, Y. Chen, L. Troup, A. Smith, and J. Granderson, "Empirical analysis of the prevalence of HVAC faults in commercial buildings," Science and Technology for the Built Environment, vol. 29, no. 10, pp. 1027–1038, 2023.

[5] W. M. P. van der Aalst, Process Mining: Data Science in Action, 2nd ed. Springer, 2016.

[6] J. Schein, S. T. Bushby, N. S. Castro, and J. M. House, "A rule-based fault detection method for air handling units," Energy and Buildings, vol. 38, no. 12, pp. 1485–1492, 2006.

[7] ASHRAE, "ASHRAE Guideline 36-2021: High-performance sequences of operation for HVAC systems," American Society of Heating, Refrigerating and Air-Conditioning Engineers, Peachtree Corners, GA, 2021. [Online]. Available: https://www.ashrae.org/technical-resources/ashrae-standards-and-guidelines

[8] ——, "ASHRAE Guideline 36-2024: High-performance sequences of operation for HVAC systems," American Society of Heating, Refrigerating and Air-Conditioning Engineers, Peachtree Corners, GA, 2024.

[9] J. Granderson, R. Singla, E. Mayhorn, P. Ehrlich, D. Vrabie, and S. Frank, "Characterization and survey of automated fault detection and diagnostic tools," Lawrence Berkeley National Laboratory, Berkeley, CA, Tech. Rep. LBNL-2001075, Nov. 2017. [Online]. Available: https://eta-publications.lbl.gov/sites/default/files/lbnl-2001075.pdf

[10] K. Heimar Andersen, S. Pommerencke Melgaard, H. Johra, A. Marszal-Pomianowska, R. Lund Jensen, and P. Kvols Heiselberg, "Barriers and drivers for implementation of automatic fault detection and diagnosis in buildings and HVAC systems: An outlook from industry experts," Energy and Buildings, vol. 303, p. 113801, 2024.

[11] X. Zhou, C. Battisti, R. Singla, G. LaPalme, G. Paliaga, and B. Eubanks, "Field demonstration of ASHRAE Guideline 36 high-performance sequences of operation for HVAC systems," Slipstream and TRC, for the Minnesota Department of Commerce, Conservation Applied Research and Development program, Tech. Rep., 2025. [Online]. Available: https://mn.gov/commerce-stat/energy/cards-2/187281/187281_slipstream_ashrae_guideline36_final_report_ada.pdf

[12] C. Battisti, Y. Li, A. Faramarzi, R. Singla, M. Mutmansky, M. Greene, and T. Jayarathne, "2028 California Energy Code CASE report: Nonresidential HVAC fault detection and diagnostics (draft)," California Statewide Codes and Standards Enhancement Team, Tech. Rep., 2026. [Online]. Available: https://title24stakeholders.com/wp-content/uploads/2026/03/2028_CASE-Report_Nonresidential-HVAC-Fault-Detection-and-Diagnostics.pdf

[13] A. R. Diallo, L. Homri, T. Boeuf, J.-Y. Dantan, and F. Bonnet, "Quantifying and mitigating alarm fatigue caused by fault detection systems," Reliability Engineering & System Safety, vol. 267, p. 111890, 2026.

[14] G. Lin, M. Pritoni, Y. Chen, and J. Granderson, "Development and implementation of fault-correction algorithms in fault detection and diagnostics tools," Energies, vol. 13, no. 10, p. 2598, 2020.

[15] M. Pritoni, G. Lin, Y. Chen, R. Vitti, C. Weyandt, and J. Granderson, "From fault-detection to automated fault correction: A field study," Building and Environment, vol. 214, p. 108900, 2022.

[16] G. Lin, J. House, Y. Chen, J. Granderson, and H. Zhang, "Active multi-mode data analysis to improve fault diagnosis in AHUs," Energy and Buildings, vol. 337, p. 115621, 2025.

[17] A. Parsaei, A. A. Markus, B. Gunay, W. O'Brien, R. Moromisato, and J. Bursill, "A proactive fault detection and diagnostics method for zone-level auto-commissioning in VAV AHUs," Energy and Buildings, vol. 344, p. 116027, 2025.

[18] Z. Wang, Y. Qin, Y. Kong, L. Wang, Q. Leng, and C. Zhang, "Advanced fault detection, diagnosis and prognosis in HVAC systems: Lifecycle insight, key challenges, and promising approaches," Renewable and Sustainable Energy Reviews, vol. 219, p. 115867, 2025.

[19] W. Tun, J. K.-W. Wong, and S.-H. Ling, "Hybrid random forest and support vector machine modeling for HVAC fault detection and diagnosis," Sensors, vol. 21, no. 24, p. 8163, 2021.

[20] Prince, B. Yoon, and P. Kumar, "Fault detection and diagnosis in air-handling unit (AHU) using improved hybrid 1D convolutional neural network," Systems, vol. 13, no. 5, p. 330, 2025.

[21] Q. Deng, Z. Chen, W. Zhu, Z. Li, Y. Yuan, and Y. Wang, "Adaptive fusion graph convolutional network based interpretable fault diagnosis method for HVAC systems enhanced by unlabeled data," Energy and Buildings, vol. 324, p. 114901, 2024.

[22] H. Wang, W. Yan, Y. Zhang, J. Yu, W. Gao, B. Zhou, and Y. Liu, "xMHAC-PDCNN: An explainable deep network with spatial feature embedding for fault detection and diagnosis of chillers under variable operating conditions," Building and Environment, vol. 284, p. 113504, 2025.

[23] K.-S. Park, J.-H. Kim, and C. S. Park, "Explainable fault detection and diagnosis in HVAC systems using a Koopman operator with normal data," Energy and Buildings, vol. 350, p. 116580, 2026.

[24] M. Paolini, M. S. Piscitelli, and A. Capozzoli, "A label-free hybrid fault detection and diagnosis approach for HVAC systems using Bayesian networks," Energy and Buildings, vol. 351, p. 116658, 2026.

[25] M. A. F. Abdollah, R. Scoccia, and M. Aprile, "Transformer encoder based self-supervised learning for HVAC fault detection with unlabeled data," Building and Environment, vol. 258, p. 111568, 2024.

[26] K. El Mokhtari and J. J. McArthur, "Autoencoder-based fault detection using building automation system data," Advanced Engineering Informatics, vol. 62, p. 102810, 2024.

[27] X. Zhu, K. Chen, B. Anduv, X. Jin, and Z. Du, "Transfer learning based methodology for migration and application of fault detection and diagnosis between building chillers for improving energy efficiency," Building and Environment, vol. 200, p. 107957, 2021.

[28] N. Ghalamsiah, J. Wen, K. S. Candan, T. Wu, Z. O'Neill, and A. Aghaei, "Unsupervised domain adaptation for HVAC fault diagnosis using contrastive adaptation network," Energy and Buildings, vol. 337, p. 115659, 2025.

[29] S. Jiang, "Leakage-safe cross-domain AHU fault detection for building operation: adaptation reliability under limited target information," Energy and Buildings, vol. 372, p. 118275, 2026.

[30] S. P. Melgaard, R. L. Jensen, P. Ferreira, K. H. Andersen, N. Dionísio, V. V. Cogo, and P. Heiselberg, "Fault detection in AHU: A walkthrough for implementation in a Danish educational building," Energy and Buildings, vol. 342, p. 115844, 2025.

[31] A. Mukhtar, M. Hadwiger, F. Wotawa, and G. Schweiger, "Reproducibility of machine learning-based fault detection and

diagnosis for HVAC systems in buildings: An empirical study," Energy and AI, vol. 22, p. 100658, 2025.
[32] S. Lin and C. Hua, "Physics-informed large language models for HVAC anomaly detection with autonomous rule generation," arXiv preprint arXiv:2510.17146, 2025. [Online]. Available: https://arxiv.org/abs/2510.17146
[33] A. Neubauer, T. Hong, H. Li, M. Yu, A. Darbandi, Y. Fürst, and M. Kriegel, "Large language models for HVAC operations in building energy systems: A critical review of methods, applications, and deployment readiness," arXiv preprint arXiv:2609.05314, 2026. [Online]. Available: https://arxiv.org/abs/2609.05314
[34] S. J. J. Leemans, D. Fahland, and W. M. P. van der Aalst, "Discovering block-structured process models from event logs — a constructive approach," in Application and Theory of Petri Nets and Concurrency (PETRI NETS 2013), ser. Lecture Notes in Computer Science, vol. 7927. Springer, 2013, pp. 311–329.
[35] F. Bezerra and J. Wainer, "Algorithms for anomaly detection of traces in logs of process aware information systems," Information Systems, vol. 38, no. 1, pp. 33–44, 2013.
[36] D. Janssen, F. Mannhardt, A. Koschmider, and S. J. van Zelst, "Process model discovery from sensor event data," in Process Mining Workshops (ICPM 2020), ser. Lecture Notes in Business Information Processing, vol. 406. Springer, 2021, pp. 69–81.
[37] E. Brzychczy, M. Aleknonytė-Resch, D. Janssen, and A. Koschmider, "Process mining on sensor data: a review of related works," Knowledge and Information Systems, vol. 67, pp. 4915–4948, 2025.
[38] T. Tratnjek, M. Kocbek Bule, and G. Polančič, "From signals to events: A maturity-based perspective on IoT-enabled process mining," Sensors, vol. 26, no. 19, p. 6293, 2026.
[39] C. Imenkamp, H. Kirchmann, S. Plumhoff, M. Weidlich, and A. Koschmider, "Glass-box activity discovery from raw sensor data," BP-Meet-IoT 2026 workshop, research track, 2026. [Online]. Available: https://bp-meet-iot.github.io/papers/bp-meet-iot2026-activity-discovery.pdf
[40] Y. Shi, N. Zhang, X. Song, H. Li, and Q. Zhu, "Novel approach for industrial process anomaly detection based on process mining," Journal of Process Control, vol. 136, p. 103165, 2024.
[41] A. Norouzifar, A. Neshastegaran, and I. Izadi, "Modeling alarm sequences using process mining techniques," in 2026 34th International Conference on Electrical Engineering (ICEE), 2026, pp. 1428–1433.
[42] F. Vitale, S. Guarino, F. Flammini, L. Faramondi, N. Mazzocca, and R. Setola, "Process mining for digital twin development of industrial cyber-physical systems," IEEE Transactions on Industrial Informatics, vol. 21, pp. 866–875, 2025.
[43] F. Vitale, N. Dall'Ora, S. Gaiardelli, E. Fraccaroli, N. Mazzocca, and F. Fummi, "Process mining-driven modeling and simulation to enhance fault diagnosis in cyber-physical systems," Journal of Manufacturing Systems, vol. 84, pp. 189–206, 2026.
[44] F. Vitale, M. Pegoraro, W. M. P. van der Aalst, and N. Mazzocca, "Control-flow anomaly detection by process mining-based feature extraction and dimensionality reduction," Knowledge-Based Systems, vol. 310, p. 112970, 2025.
[45] E. J. Husom, A. Goknil, F. Mannhardt, S. Tverdal, S. Sen, and P. Nguyen, "Unsupervised learning and process analysis for sensor data validation in the IIoT," ACM Transactions on Internet Technology, vol. 26, no. 2, pp. 1–34, 2026.
[46] A. Moradbeikie, A. P. A. d. C. Barbon, I. M. Grigore, D. F. Barbin, and S. Barbon Junior, "Process mining of sensor data for predictive process monitoring: A HACCP-guided pasteurization study case," Systems, vol. 13, no. 11, p. 935, 2025.
[47] W. Deabes, K. E. Bouazza, and W. Algthami, "Smart fuzzy Petri net-based temperature control framework for reducing building energy consumption," Sensors, vol. 23, no. 13, p. 5985, 2023.
[48] K. Akhramovich, E. Serral, and C. Cetina, "A systematic literature review on the application of process mining to Industry 4.0," Knowledge and Information Systems, vol. 66, no. 5, pp. 2699–2746, 2024.
[49] Z. Khais Shahid, S. Saguna, and C. Åhlund, "Unsupervised analysis of EcoStruxure Building Operation (EBO) alarm logs in school buildings," in Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '26), 2026, pp. 282–292.
[50] J. Granderson, G. Lin, Y. Chen, A. Casillas, J. Wen, Z. Chen, P. Im, S. Huang, and J. Ling, "A labeled dataset for building HVAC systems operating in faulted and fault-free states," Scientific Data, vol. 10, p. 342, 2023.
[51] SES Consulting Inc., "Thompson Rivers University ASHRAE Level 1 energy study: Campus Activity Centre," Prepared for Thompson Rivers University, Vancouver, BC, Tech. Rep., May 2022. [Online]. Available: https://www.tru.ca/___shared/assets/level-1-energy-study-2022-cac64231.pdf